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Multi-slot die coater

Abstract

A multi-slot die coater includes lower and upper slots; lower, intermediate, and upper die blocks; and first and second manifolds. The first and second manifold are provided in the lower and intermediate die blocks, respectively, to communicate with the respective lower and upper slots. A first bolt is fastened to a contact surface between the upper die block and the intermediate die block, and a distance from an intermediate die lip to one end of the second manifold is A, a distance between one end of the second manifold and the other end is B, and a distance from the other end of the second manifold to an axis of the first bolt is C in an order of how far it is away from the intermediate die lip, $C/(A+B+C)$ is 28% or less.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) The present application is a national phase entry under 35 U.S.C. § 371 of International Application No. PCT/KR2021/011400, filed on Aug. 25, 2021, which claims priority to Korean Patent Application No. 10-2020-0125182, filed on Sep. 25, 2020, the disclosures of which are incorporated herein by reference.

TECHNICAL FIELD

(2) The present disclosure relates to a multi-slot die coater capable of simultaneously forming two or more layers by wetting, and more particularly, to a multi-slot die coater with suppressed deformation and twist which are induced by the structural feature due to including a thin die.

BACKGROUND ART

(3) With the increasing technology development and the growing demand for mobile devices, the demand for secondary batteries as an energy source is rapidly increasing, and such secondary batteries essentially include an electrode assembly which is a power generation element. The electrode assembly includes a positive electrode, a separator and a negative electrode stacked at least once, and the positive electrode and the negative electrode are manufactured by coating and drying a positive electrode active material slurry and a negative electrode active material slurry on a current collector made of an aluminum foil and a current collector made of a copper foil, respectively. For the uniform charging/discharging characteristics of the secondary batteries, it is necessary to uniformly coat the positive electrode active material slurry and the negative electrode active material slurry on the current collector, and slot die coaters have been used.

(4) The electrode manufacturing method using the slot die coater includes applying the electrode active material slurry issuing from the slot die coater to the current collector transferred by a coating roll. The conventional slot die coater includes two dies and a slot between the two dies, and may dispense one type of electrode active material slurry through one slot to form a layer of electrode active material.

(5) To manufacture the secondary batteries with high energy density, the thickness of the electrode active material layer which was about 130 μm has gradually increased up to 300 μm . When the thick electrode active material layer is formed by the conventional slot die coater, a binder and a conductive material in the active material slurry migrate fast during drying, and the finally manufactured electrode is not uniform. To solve this problem, when forming the electrode active material layer at a small thickness and drying and repeating this process thereon, it takes a long time to perform coating twice. To improve both electrode performance and productivity, the inventors have proposed a dual slot die coater capable of simultaneously applying two types of electrode active material slurries.

(6) FIG. 1 is a schematic cross-sectional view of the dual slot die coater according to the related art.

(7) Referring to FIG. 1, two electrode active material layers may be simultaneously formed on a current collector **15** by applying two types of electrode active material slurries from the dual slot die coater **20** while rotating a coating roll **10** to move the current collector **15**. The electrode active material slurry coming out of the dual slot die coater **20** is applied over one surface of the current

collector **15** to form the electrode active material layer.

(8) The dual slot die coater **20** is constructed by assembling three plate members, i.e., three die blocks **21**, **22**, **23**. Slots are formed between the adjacent die blocks, and thus two slots are formed, and the two types of electrode active material slurries are simultaneously dispensed through exit ports **24**, **25**, each in communication with each slot, so two electrode active material layers may be simultaneously formed by continuously applying an additional electrode active material slurry on the electrode active material layer formed by the previously coated electrode active material slurry. The reference numbers **26**, **27** indicate manifolds in which a coating solution is received.

(9) However, due to using the electrode active material slurries simultaneously issuing from the different exit ports **24** and **25**, the process using the dual slot die coater **20** is quite difficult to form each electrode active material layer at a desired thickness.

(10) In general, the thickness of each electrode active material layer is affected by the amount of the electrode active material slurry coming out through the exit ports **24**, **25**, and the amount of the electrode active material slurry coming out is affected by the size of each exit port **24**, **25** (slot gap), and thus to form a desired thickness, the conventional art needs to repeat the task of testing the coating process a few times, disassembling and re-assembling the die blocks **21**, **22**, **23**, adjusting the slot gap and checking the amount of the electrode active material slurry coming out. However, the slot gap is a variable that is adjusted so sensitively that the slot gap greatly changes depending on the fastening strength of bolts used to assemble the die blocks **21**, **22**, **23**, and may be changed by a force used to pump the electrode active material slurry. In particular, to stably perform uniform coating in the widthwise direction (TD direction) along the movement direction (MD direction) of the current collector, uniform dimensional precision in the widthwise direction is needed, and it is more difficult to uniformly control the widthwise slot gap as the width of the dual slot die coater **20** increases to use the wide current collector in order to increase the production amount.

(11) Since the slot die coater **20** basically includes three die blocks **21**, **22**, **23**, to configure a device having a foot print and volume similar to the conventional slot die coater including one slot, it is necessary to reduce the thickness of each of the die blocks **21**, **22**, **23**, and by this reason, inevitably, it is structurally vulnerable to deformation and twist. When deformation or twist occurs, the adjusted slot gap is changed, causing defects in the electrode process. Furthermore, this problem will be more serious with a multi-slot die coater including two or more slots due to the increased number of die blocks.

(12) To solve this problem, when each die block **21**, **22**, **23** increases in size (change in angle), the direction in which the slurry emerges changes, resulting in reduced coating workability.

Additionally, among the three die blocks **21**, **22**, **23**, when the die blocks **21**, **23** disposed on the outer side increases in thickness to prevent deformation and twist, it is still difficult to prevent deformation of the most structurally vulnerable intermediate die block **22**.

(13) Meanwhile, each die block **21**, **22**, **23** may be assembled by bolt fastening to the contact surface. However, each die block **21**, **22**, **23** is vulnerable to deformation due to its small thickness, and bolt fastening is prone to cause deformation by the action of a large force, so caution is needed. When movement is made between the die blocks **21**, **22**, **23** by the force applied when fastening, the movement affects the coating gap, i.e., the distance between the exit ports **24**, **25** and the current collector **15**, causing loading deviation, resulting in ununiform coating.

DISCLOSURE

Technical Problem

(14) The present disclosure is designed to solve the above-described problem, and therefore the present disclosure is directed to providing a multi-slot die coater with reduced structural vulnerability to deformation and twist in the multi-slot die coater basically including three die blocks.

(15) However, the technical problems to be solved by the present disclosure are not limited to the

above problems, and other problems that are not mentioned will be clearly understood by those skilled in the art from the description of the invention described below.

Technical Solution

(16) To solve the above-described technical problem, a multi-slot die coater of the present disclosure is a multi-slot die coater including a lower slot and an upper slot to extrude and coat a coating solution on a surface of a continuously moving substrate through at least one of the lower slot or the upper slot, and the multi-slot die coater includes a lower die block, an intermediate die block positioned on the lower die block such that the lower slot is formed between the intermediate die block and the lower die block, and an upper die block positioned on the intermediate die block such that the upper slot is formed between the upper die block and the intermediate die block; a first manifold provided in the lower die block, the first manifold in which a first coating solution is received, the first manifold being in communication with the lower slot, and a second manifold provided in the intermediate die block, the second manifold in which a second coating solution is received, the second manifold being in communication with the upper slot; and a first bolt fastened to a contact surface between the upper die block and the intermediate die block, wherein the lower die block, the intermediate die block and the upper die block include a lower die lip, an intermediate die lip and an upper die lip that form front ends thereof, respectively, and when a distance from the intermediate die lip to one end of the second manifold is A, a distance between one end of the second manifold and the other end is B, and a distance from the other end of the second manifold to an axis of the first bolt is C in an order of how far it is away from the intermediate die lip, $C/(A+B+C)$ is 28% or less.

(17) The first bolt may be vertically fastened to the contact surface between the upper die block and the intermediate die block.

(18) The vertical range may be an angle range of $90\pm 10^\circ$ from the contact surface.

(19) The contact surface between the intermediate die block and the upper die block may be parallel to a horizontal plane.

(20) The lower slot and the upper slot may form an angle of 30° to 60° .

(21) In the present disclosure, the multi-slot die coater may further include a second bolt fastened to the contact surface between the intermediate die block and the lower die block, wherein when a distance from the lower die lip to one end of the first manifold is A', a distance between one end of the first manifold and the other end is B', and a distance from the other end of the first manifold to an axis of the second bolt is C' in an order of how far it is away from the lower die lip, $C'/(A'+B'+C')$ is 28% or less.

(22) The second bolt may be vertically fastened to the contact surface between the intermediate die block and the lower die block.

(23) The contact surface between the lower die block and the intermediate die block may be inclined with respect to a horizontal plane.

(24) A lower exit port may be formed between the lower die lip and the intermediate die lip and may be in communication with the lower slot, an upper exit port may be formed between the intermediate die lip and the upper die lip and may be in communication with the upper slot, and a bolt head of the second bolt may be farther away from the lower exit port and the upper exit port than a bolt head of the first bolt.

Advantageous Effects

(25) It is possible to minimize deformation of the die blocks by positioning the bolts according to the present disclosure. Thus, it is possible to reduce a change in coating gap when mounting the multi-slot die coater by assembling the die blocks. It is also possible to suppress slot gap deviation. Accordingly, according to the present disclosure, it is possible to ensure widthwise loading uniformity, thereby improving the coating quality.

(26) According to the present disclosure, it is possible to reduce the structural vulnerability of the die block to deformation or twist, and thereby uniformly form the coating layer, especially, the

electrode active material layer, at a desired thickness, and it is possible to coat at least two types of electrode active material slurries at the same time, thereby achieving high performance and productivity. It is possible to prevent deformation when fastening the die blocks, thereby ensuring coating workability and reproducibility. When the multi-slot die coater of the present disclosure is used to coat an electrode active material slurry on a current collector in the manufacture of an electrode of a secondary battery, it is possible to achieve uniform coating under high-speed or wide-scale coating conditions.

Description

DESCRIPTION OF DRAWINGS

- (1) The accompanying drawings illustrate a preferred embodiment of the present disclosure and together with the detailed description of the present disclosure, serve to provide further understanding of the technical features of the present disclosure, and thus, the present disclosure is not construed as being limited to the drawing.
- (2) FIG. 1 is a schematic cross-sectional view of a dual slot die coater according to the related art.
- (3) FIG. 2 is a schematic cross-sectional view of a multi-slot die coater according to an embodiment of the present disclosure.
- (4) FIG. 3 is a schematic exploded perspective view of a multi-slot die coater according to an embodiment of the present disclosure.
- (5) FIG. 4 is a cross-sectional view showing bolt fastening between die blocks in a multi-slot die coater according to an embodiment of the present disclosure.

BEST MODE

- (6) Hereinafter, preferred embodiments of the present disclosure will be described in detail with reference to the accompanying drawings. Prior to the description, it should be understood that the terms or words used in the specification and the appended claims should not be construed as limited to general and dictionary meanings, but interpreted based on the meanings and concepts corresponding to technical aspects of the present disclosure on the basis of the principle that the inventor is allowed to define terms appropriately for the best explanation. Therefore, the embodiments described herein and illustrations in the drawings are just some preferred embodiments of the present disclosure and do not fully describe the technical features of the present disclosure, so it should be understood that a variety of other equivalents and modifications could have been made thereto at the time of filing the patent application.
- (7) A multi-slot die coater of the present disclosure may include two or more slots. Basically, the multi-slot die coater is an apparatus including a lower slot and an upper slot to coat a coating solution in a double layer on a substrate. The 'substrate' described below is a current collector and the coating solution is an 'electrode active material slurry'. Both a first coating solution and a second coating solution are electrode active material slurries, and they may have the same or different compositions (types of an active material, a conductive material and a binder), amounts (amounts of the active material, the conductive material and the binder) or properties. The multi-slot die coater of the present disclosure is optimized for electrodes manufactured by applying at least two types of electrode active material slurries at the same time or by pattern coating of applying at least two types of electrode active material slurries in an alternating manner. However, the scope of the present disclosure is not necessarily limited thereto. For example, the substrate may be a porous support for a separator, and the first coating solution and the second coating solution may be organics having different compositions or properties. That is, in case that thin film coating is required, the substrate, the first coating solution and the second coating solution are not limited to particular types.
- (8) The inventors found that the die blocks are deformed by the position of bolts used to fasten the

die blocks, and arrived at the present disclosure. Since the die blocks are deformed by the bolt position, the inventors propose the present disclosure based on the idea that to prevent a change in slot gap, bolt positioning is important. The present disclosure proposes the position of bolts used to fasten the die blocks to minimize the loading deviation.

(9) FIG. 2 is a schematic cross-sectional view of the multi-slot die coater according to an embodiment of the present disclosure. FIG. 3 is a schematic exploded perspective view of the multi-slot die coater according to an embodiment of the present disclosure.

(10) The multi-slot die coater **100** according to the present disclosure is a dual slot die coater including a lower slot **101** and an upper slot **102**, and is an apparatus capable of simultaneously or alternately coating a same type of coating solution or two different types of coating solutions on a substrate **300** through the lower slot **101** and the upper slot **102**. Referring to FIGS. 2 and 3, the multi-slot die coater **100** includes a lower die block **110**, an intermediate die block **120** positioned on the lower die block **110**, and an upper die block **130** positioned on the intermediate die block **120**. The die blocks **110**, **120**, **130** are assembled with one another through a first bolt **140** and a second bolt **150**.

(11) The lower die block **110** is the lowermost block among the blocks of the multi-slot die coater **100**, and the surface facing the intermediate die block **120** is inclined at an angle of approximately 30° to 60° to the bottom surface (X-Z plane). The intermediate die block **120** is positioned in contact with the upper surface of the lower die block **110** inclined with respect to the horizontal plane. The lower slot **101** is formed at a location in which the intermediate die block **120** and the lower die block **110** are in contact with each other.

(12) For example, a first spacer **113** is interposed between the lower die block **110** and the intermediate die block **120** to form a gap between, and the lower slot **101** corresponding to a passage through which the first coating solution **50** may flow may be formed. In this case, the thickness of the first spacer **113** determines the vertical width (Y-axis direction, the slot gap) of the lower slot **101**.

(13) As shown in FIG. 3, the first spacer **113** has a first opening portion **113a** which is cut at an area, and may be interposed in the remaining portion except one side in the edge area of the facing surface of each of the lower die block **110** and the intermediate die block **120**. Accordingly, a lower exit port **101a** through which the first coating solution **50** emerges is only formed between the front end of the lower die block **110** and the front end of the intermediate die block **120**. The front end of the lower die block **110** and the front end of the intermediate die block **120** are defined as a lower die lip **111** and an intermediate die lip **121**, respectively, and in other words, the lower exit port **101a** is formed by the spacing between the lower die lip **111** and the intermediate die lip **121**.

(14) For reference, the first spacer **113** acts as a gasket to prevent the leakage of the first coating solution **50** through the gap between the lower die block **110** and the intermediate die block **120** except the area where the lower exit port **101a** is formed, and thus the first spacer **113** is preferably made of a material having sealing ability.

(15) The lower die block **110** includes a first manifold **112** having a predetermined depth on the surface facing the intermediate die block **120**, and the first manifold **112** is in communication with the lower slot **101**. Although not shown in the drawings, the first manifold **112** is connected to a first coating solution supply chamber (not shown) installed outside with a supply pipe and is supplied with the first coating solution **50**. When the first manifold **112** is fully filled with the first coating solution **50**, the flow of the first coating solution **50** is guided along the lower slot **101** and comes out of the lower exit port **101a**.

(16) The intermediate die block **120** is a block in the middle of the blocks of the multi-slot die coater **100**, and is a block interposed between the lower die block **110** and the upper die block **130** to form a dual slot. The intermediate die block **120** of this embodiment is a right-angled triangle in cross section. According to another embodiment, the intermediate die block **120** may be, for example, an isosceles triangle in cross section.

(17) The upper die block **130** is positioned in contact with the upper surface of the intermediate die block **120** parallel to the bottom surface. The upper slot **102** is formed at a location in which the intermediate die block **120** and the upper die block **130** are in contact with each other.

(18) In the same way as the lower slot **101** described above, a second spacer **133** may be interposed between the intermediate die block **120** and the upper die block **130** to form a gap between. Accordingly, the upper slot **102** corresponding to a passage through which a second coating solution **60** may flow is formed. In this case, the vertical width (Y-axis direction, the slot gap) of the upper slot **102** is determined by the second spacer **133**.

(19) In addition, the second spacer **133** having the similar structure to the first spacer **113** has a second opening portion **133a** which is cut at an area, and is interposed in the remaining portion except one side in the edge area of the facing surface of each of the intermediate die block **120** and the upper die block **130**. Likewise, the circumferential direction except the front side of the upper slot **102** is blocked, and the upper exit port **102a** is only formed between the front end of the intermediate die block **120** and the front end of the upper die block **130**. The front end of the upper die block **130** is defined as an upper die lip **131**, and in other words, the upper exit port **102a** is formed by the spacing between the intermediate die lip **121** and the upper die lip **131**.

(20) In addition, the intermediate die block **120** includes a second manifold **132** having a predetermined depth on the surface facing the intermediate die block **120**, and the second manifold **132** is in communication with the upper slot **102**. Although not shown in the drawings, the second manifold **132** is connected to a second coating solution supply chamber installed outside with a supply pipe and is supplied with the second coating solution **60**. When the second coating solution **60** is supplied from the external source along the supply pipe, and the second manifold **132** is fully filled with the second coating solution **60**, the flow of the second coating solution **60** is guided along the upper slot **102** in communication with the second manifold **132** and comes out of the upper exit port **102a**.

(21) The upper slot **102** and the lower slot **101** form an angle, and the angle may be approximately 30° to 60°. The upper slot **102** and the lower slot **101** may intersect at one point, and the upper exit port **102a** and the lower exit port **101a** may be provided near the intersection point. Accordingly, the locations at which the first coating solution **50** and the second coating solution **60** emerge may be concentrated on approximately one point.

(22) In this embodiment, the first manifold **112** is formed in the lower die block **110**, and the second manifold **132** is formed in the intermediate die block **120**. The first manifold **112** may be formed in the intermediate die block **120**, and the second manifold **132** may be formed in the upper die block **130**.

(23) According to the multi-slot die coater **100** having such a configuration, a rotatable coating roll **200** is positioned in front of the multi-slot die coater **100**, and the coating roll **200** may be rotated to move the substrate **300** to be coated, while continuously contacting the first coating solution **50** and the second coating solution **60** with the surface of the substrate **300**, and thereby the substrate **300** may be coated in a double layer. Alternatively, pattern coating may be intermittently formed on the substrate **300** by performing the supply and stop of the first coating solution **50** and the supply and stop of the second coating solution **60** in an alternating manner.

(24) The multi-slot die coater **100** extrudes and coats the coating solution on the surface of the continuously moving substrate **300** through at least one of the lower slot **101** or the upper slot **102**. In this embodiment, the contact surface between the intermediate die block **120** and the upper die block **130** is parallel to the horizontal plane. The first bolt **140** is fastened to the contact surface. Additionally, the contact surface between the intermediate die block **120** and the lower die block **110** is inclined with respect to the horizontal plane. The second bolt **150** is fastened to the contact surface. There may be at least one first bolt **140** and at least one second bolt **150**. FIG. 3 shows an example in which the first bolt **140** is inserted in the downward direction from the rear surface of the upper die block **130**, and a plurality of first bolts **140** is fastened along the widthwise direction

of the upper die block **130**. The second bolt **150** or the first bolt **140** is characterized in having the following positions. It will be described in detail.

(25) FIG. **4** is a cross-sectional view showing the bolt fastening between the die blocks in the multi-slot die coater according to an embodiment of the present disclosure.

(26) In the order of how far it is away from the intermediate die lip **121**, when the distance from the intermediate die lip **121** to one end of the second manifold **132** is A, the distance between one end of the second manifold **132** and the other end is B, and the distance from the other end of the second manifold **132** to the axis of the first bolt **140** is C, $C/(A+B+C)$ is 28% or less.

(27) Here, an area from the intermediate die lip **121** to one end of the second manifold **132** is referred to as land. The area between one end of the second manifold **132** and the other end corresponds to the size when viewed from the second manifold **132**. The second coating solution **60** fully filled in the second manifold **132** flows from the second manifold **132** via the upper slot **102** and is coated on the substrate **300** through the upper exit port **102a**. Accordingly, the second coating solution **60** is present from the other end of the second manifold **132** to the intermediate die lip **121**. Additionally, the length of the area in which the second coating solution **60** exists is A+B. The distance C from the other end of the second manifold **132** to the axis of the first bolt **140** defines the fastening position of the first bolt **140**.

(28) When the first bolt **140** is too close to the intermediate die lip **121**, a slurry coating area (an area in which the second coating solution **60** exists, an area from the second manifold **132** to the upper exit port **102a**, its length is A+B) reduces and the loading deviation increases. When the first bolt **140** is too far away from the intermediate die lip **121**, deformation occurs in the opposite slot gap by the adverse effect of the first bolt **140** tightening (leverage effect), causing greater loading deviation. Accordingly, it is necessary to fasten the first bolt **140** to an appropriate position from the intermediate die lip **121**. The present disclosure proposes a new parameter $C/(A+B+C)$ of 28% or less in setting the appropriate fastening position. When A, B, C have this relationship, it is possible to minimize the loading deviation. $C/(A+B+C)$ is 28% or less. The upper limit is important, and the lower limit does not matter. When $C/(A+B+C)$ is more than 28%, deformation occurs in the opposite slot gap by the adverse effect of the first bolt **140** tightening, causing greater loading deviation, and accordingly, $C/(A+B+C)$ is 28% or less. As the first bolt **140** is closer to the other end of the second manifold **132**, C gets close to 0. However, since the first bolt **140** is not fastened into the second manifold **132** (leakage), C is larger than 0. A and B are determined by the design of the intermediate die block **120** and the second manifold **132**.

(29) The first bolt **140** may be vertically fastened to the contact surface between the upper die block **130** and the intermediate die block **120**. The vertical range may be an angle range of $90\pm 10^\circ$ from the contact surface.

(30) The fastening position of the second bolt **150** may be the same as the fastening position setting of the first bolt **140** as described above. That is, in the order of how far it is away from the lower die lip **111**, when the distance from the lower die lip **111** to one end of the first manifold **112** is A', the distance between one end of the first manifold **112** and the other end is B', and the distance from the other end of the first manifold **112** to the axis of the second bolt **150** is C', $C'/(A'+B'+C')$ is 28% or less. When A', B', C' have this relationship, it is possible to minimize the loading deviation by the above-described reason.

(31) Meanwhile, the bolt head of the second bolt **150** may be farther away from the lower exit port **101a** and the upper exit port **102a** than the bolt head of the first bolt **140**. By this configuration, it is possible to prevent deformation of the die blocks **110**, **120**, **130** through uniform distribution of the fastening force without interference between the first bolt **140** and the second bolt **150** while not affecting the first manifold **112** and the second manifold **132**.

(32) The second bolt **150** may be vertically fastened to the contact surface between the intermediate die block **120** and the lower die block **110**. The vertical range may be an angle range of $90\pm 10^\circ$ from the contact surface.

(33) When the second bolt **150** is fastened, at an angle, not vertically, to the inclined contact surface between the intermediate die block **120** and the lower die block **110**, forces by the second bolt **150** tightening are split into force perpendicular to the contact surface and force parallel to the contact surface. Thus, when fastening, the intermediate die block **120** will slip by the force parallel to the contact surface. When movement is made between the die blocks **110**, **120**, **130** by the force applied when fastening, the movement affects the coating gap, i.e. the distance between the lower and upper exit ports **101a**, **102a** and the substrate **300**, causing ununiform coating. Accordingly, it is desirable to vertically fasten the second bolt **150** to the inclined contact surface. When the second bolt **150** is vertically fastened to the inclined contact surface, the direction of the force when fastening the second bolt **150** is perpendicular to the inclined contact surface and slip does not occur. Accordingly, in the second bolt **150** which does not simply fix, and applies the fastening torque, the tightening force of the second bolt **150** prevents the slip of the intermediate die block **120** on the upper die block **110**. Accordingly, it is possible to reduce a change in coating gap when mounting the multi-slot die coater **100** by assembling the die blocks **110**, **120**, **130**.

(34) As described above, according to the present disclosure, it is possible to minimize deformation of the die blocks **110**, **120**, **130** and ensure widthwise loading uniformity, thereby improving the coating quality. Additionally, according to the present disclosure, it is possible to suppress the slot gap deviation.

(35) Although this embodiment describes applying the coating solution in two layers or performing pattern coating by supplying the coating solution in an alternating manner, it is obvious that the present disclosure may be applied to mixing two types of coating solutions in the middle of the slot, not dispensing through individual slots, or simultaneous coating of three or more layers using three or more slots. It is obvious that four or more die blocks are needed for three or more slots.

(36) Hereinafter, an example of demonstrating the effect of the present disclosure through the simulation results will be described. The internal flow simulation of the multi-slot die coater is carried out.

(37) The loading deviation refers to flow rate deviation at the end of the die lip (maximum value (MAX) of flow rate-minimum value (MIN) of flow rate). For uniform coating quality, it is necessary to control the loading deviation below 10%. The loading deviation simulation is carried out by variously combining A, B, C. The current simulation technology is excluded since it is difficult to accurately interpret the external flow. Since the influence of the external flow of the die blocks may be reflected, there may be a slight difference from the loading deviation in the actual multi-slot die coater.

(38) Table 1 summarizes A, B, C conditions of example and comparative example and their loading deviation.

(39) TABLE-US-00001 TABLE 1 Loading deviation (MAX- A B C C/(A + B) C/(A + B + C) MIN)% Example 1 50 50 17 17.00% 14.53% 2.3 Example 2 50 50 20 20.00% 16.67% 2.6 Example 3 50 50 30 30.00% 23.08% 4.8 Comparative 50 50 40 40.00% 28.57% 11 example 1 Comparative 50 50 50 50.00% 33.33% 15 example 2 Comparative 50 50 60 60.00% 37.50% 21 example 3

(40) In example and comparative example, A and B are 50 (unit: a.u), and C of example increases to 17, 20, 30. C of comparative example increases to 40, 50, 60. C increases as it goes from example 1 to comparative example 3. The calculation results of $C/(A+B+C)$ are: 14.53% in example 1, 16.67% in example 2, 23.08% in example 3, 28.57% in comparative example 1, 33.33% in comparative example 2, and 37.50% in comparative example 3. Since A, B are fixed and C increases, $C/(A+B+C)$ increases. As a result of calculating the loading deviation in each case, the loading deviation of example is 2.3 to 4.8% which satisfy the management range of less than 10%. In contrast, the loading deviation of comparative example is calculated as 11 to 21%, and thus is evaluated as a level exceeding the current management range of 10%.

(41) In particular, from the findings that when $C/(A+B+C)$ of comparative example 1 is 28.57%, the loading deviation is 11%, it is concluded that $C/(A+B+C)$ satisfying the loading deviation of

less than 10% is 28% or less. $C/(A+B+C)$ is a new parameter that has never been considered, and it has a significance as a new parameter introduced to minimize the loading deviation, and at the present time, for the loading deviation of less than 10%.

(42) According to the present disclosure, it is possible to prevent deformation or twist of the die block having structural vulnerability, and thereby uniformly form the coating layer, in particular, the electrode active material layer at a desired thickness, and it is possible to coat two types of electrode active material slurries at the same time, thereby achieving high performance and productivity. It is possible to prevent slip when fastening the die blocks, thereby ensuring coating workability and reproducibility. When the multi-slot die coater of the present disclosure is used to coat an electrode active material slurry on a current collector in the manufacture of an electrode of a secondary battery, it is possible to achieve uniform coating under high-speed or wide-scale coating conditions.

(43) While the present disclosure has been described with respect to a limited number of embodiments and drawings, the present disclosure is not limited thereto, and it will be apparent to those skilled in the art that a variety of changes and modifications may be made thereto within the technical aspects of the present disclosure and the appended claims and their equivalent scope.

Claims

1. A multi-slot die coater comprising: a lower slot and an upper slot, a lower die block, an intermediate die block positioned on the lower die block such that the lower slot is formed between the intermediate die block and the lower die block, and an upper die block positioned on the intermediate die block such that the upper slot is formed between the upper die block and the intermediate die block; a first manifold provided in the lower die block, the first manifold in which a first coating solution is received, the first manifold being in communication with the lower slot, and a second manifold provided in the intermediate die block, the second manifold in which a second coating solution is received, the second manifold being in communication with the upper slot; and a first bolt fastened to a contact surface between the upper die block and the intermediate die block, wherein the lower die block, the intermediate die block and the upper die block include a lower die lip, an intermediate die lip and an upper die lip that form front ends thereof, respectively, and wherein when a distance from the intermediate die lip to one end of the second manifold is A, a distance between one end of the second manifold and the other end is B, and a distance from the other end of the second manifold to an axis of the first bolt is C in an order of how far it is away from the intermediate die lip, $C/(A+B+C)$ is 0.28 or less, wherein the front end of the lower die lip, the intermediate die lip, and the upper die lip are parallel to a vertical plane of a discharge end of the multi-slot die coater.

2. The multi-slot die coater according to claim 1, wherein the first bolt is vertically fastened to the contact surface between the upper die block and the intermediate die block.

3. The multi-slot die coater according to claim 2, wherein a vertical range is an angle range of $90 \pm 10^\circ$ from the contact surface.

4. The multi-slot die coater according to claim 1, wherein the contact surface between the intermediate die block and the upper die block is parallel to a horizontal plane.

5. The multi-slot die coater according to claim 1, wherein the lower slot and the upper slot form an angle of 30° to 60° .

6. The multi-slot die coater according to claim 1, further comprising: a second bolt fastened to the contact surface between the intermediate die block and the lower die block, wherein when a distance from the lower die lip to one end of the first manifold is A', a distance between one end of the first manifold and the other end is B', and a distance from the other end of the first manifold to an axis of the second bolt is C' in an order of how far it is away from the lower die lip, $C'/(A'+B'+C')$ is 0.28 or less.

7. The multi-slot die coater according to claim 6, wherein the second bolt is vertically fastened to the contact surface between the intermediate die block and the lower die block.
 8. The multi-slot die coater according to claim 7, wherein a vertical range is an angle range of $90\pm 10^\circ$ from the contact surface.
 9. The multi-slot die coater according to claim 7, wherein the contact surface between the lower die block and the intermediate die block is inclined with respect to a horizontal plane.
 10. The multi-slot die coater according to claim 7, wherein a lower exit port is formed between the lower die lip and the intermediate die lip and is in communication with the lower slot, an upper exit port is formed between the intermediate die lip and the upper die lip and is in communication with the upper slot, and a bolt head of the second bolt is farther away from the lower exit port and the upper exit port than a bolt head of the first bolt.
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